Introduction

lme4::lmer() is the standard R implementation of linear mixed-effects models and the de facto reference for the formula syntax used across the broader mixed-models ecosystem (lmerTest, glmmTMB, brms, nlme). Mastering the random-effects syntax — what (1 | group), (x | group), (x || group), and nested vs. crossed grouping notation actually mean — is a prerequisite for fluent mixed-models work. The output structure is equally important: each block (fixed effects, random effects, residual variance) carries different inferential weight and must be reported correctly.

Prerequisites

A working understanding of mixed-effects models, the difference between fixed and random effects, and basic familiarity with lme4 formula notation.

Theory

Core random-effects syntax in lmer():

- (1 | g): random intercept per level of g.
- (x | g): random intercept *and* correlated random slope for x within g.
- (x || g) or equivalently (1 | g) + (0 + x | g): uncorrelated random intercept and random slope.
- (1 | g/h): h nested within g (each h value belongs to one g).
- (1 | g) + (1 | h): g and h as crossed random factors.
- (0 + x | g): random slope without random intercept; rarely scientifically motivated.

The fixed-effects part of the formula (Reaction ~ Days) is interpreted as in lm(); the random-effects part adds the variance components.

Assumptions

Linearity in the linear predictor, conditional independence given random effects, multivariate Normal random effects, Normal residuals with constant variance, and independence of residuals from random effects.

R Implementation

```r
library(lme4); library(lmerTest)

data(sleepstudy, package = "lme4")

# Random intercept only
m1 <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)

# Random intercept + correlated slope
m2 <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)

# Random intercept + uncorrelated slope
m3 <- lmer(Reaction ~ Days + (1 | Subject) + (0 + Days | Subject),
           data = sleepstudy)

# Nested example (sleepstudy doesn't have nesting; use simulated)
set.seed(2026)
dat <- data.frame(g = gl(5, 20), subj = factor(1:100),
                  x = rnorm(100), y = rnorm(100))
m_nest <- lmer(y ~ x + (1 | g/subj), data = dat)
```

```r
anova(m1, m2, m3)
summary(m2)
```

Output & Results

`summary()` reports four blocks: fixed-effect coefficients with standard errors and (with `lmerTest`) Satterthwaite degrees of freedom and p-values; random-effect variance components and correlations between them; residual variance; and overall fit statistics including REML criterion. `anova()` between models compares them via likelihood-ratio tests when fitted with ML.

Interpretation

Reading `summary(m2)`:

- **Fixed effects**: population-level intercept and slope with standard errors and (with `lmerTest`) p-values.
- **Random effects**: per-subject intercept and slope variances, plus their correlation; the SD entries quantify between-subject heterogeneity.
- **Residual SD**: within-subject (after partialling out the random effects) error.
- **Number of obs / groups**: confirms the model is fitting the structure you intended.

A reporting sentence: "A random-intercept-and-slope model gave a population-level reaction-time slope of 10.5 ms per day of sleep deprivation (95 % CI 7.6 to 13.4) with substantial between-subject heterogeneity in slope (SD = 5.92, correlation with random intercept −0.04) and residual SD of 25.6 ms." Report variance components alongside fixed effects.

Practical Tips

- Use `lmerTest` for Satterthwaite or Kenward-Roger degrees of freedom; `lme4` alone returns no p-values for fixed effects.
- For convergence warnings, rescale predictors (`scale()` continuous variables) and check with `allFit()` across multiple optimisers; a stable estimate across optimisers is reassuring.
- Singular fits (correlation between random effects estimated as ± 1) usually mean the random structure is over-specified for the data; simplify by removing slope-intercept correlations or random slopes.
- For likelihood-ratio tests of variance components, the null is on the boundary; use the half-half χ^2 mixture or the parametric bootstrap (`pbkrtest::PBmodcomp`) rather than the standard χ^2.
- For non-Gaussian outcomes, `glmer()` extends the same formula syntax with `family = binomial`, `poisson`, etc.
- For Bayesian fits with identical formulas, `brms::brm()` is a near drop-in replacement and adds proper credible intervals.

R Packages Used

`lme4` for `lmer()` and `glmer()`; `lmerTest` for Satterthwaite / Kenward-Roger p-values; `pbkrtest` for parametric-bootstrap likelihood-ratio tests; `glmmTMB` for an alternative implementation that handles overdispersed and zero-inflated families; `brms` for Bayesian mixed models with the same formula syntax.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI